SUBJECT CODE : 23A01702b

RAILWAYS, AIRPORTS, DOCKS AND HARBOUR ENGINEERING

UNIT – II : GEOMETRIC DESIGN OF RAILWAY TRACK

(Handwritten Study Notes - Exam Oriented)

Syllabus (as given):

Gradients – Grade Compensation – Cant and Negative Super Elevation – Cant Deficiency – Degree of Curves – Safe Speed on Railway Track – Points and Crossings – Layout and Functioning of Left Hand Turn Out and Right Hand Turn Outs – Station Yards – Signaling and Interlocking

1. Gradients

Gradient is the rate of rise or fall of the track along its length. It is expressed as **1 in n** (1 vertical in n horizontal) or as a percentage: **% = 100 / n**. Ex: 1 in 200 = 0.5 %.

A gradient is provided to (i) follow the natural ground surface, (ii) reduce earthwork, and (iii) provide drainage in yards.

Types of gradients

Gradient	Meaning	Typical value (BG)
Ruling gradient	Maximum gradient on a section which decides the maximum load a locomotive can haul	1 in 150 – 1 in 200 (plains); 1 in 100 (hills)
Momentum gradient	Steeper than ruling; a train can climb it using the momentum gained on the preceding down gradient	Steeper than ruling, short length
Pusher / helper gradient	Steep gradient on hills where an extra (pusher) engine is needed to haul the train	1 in 37.5 (Bhor ghat); 1 in 100 on hills

Gradient	Meaning	Typical value (BG)
Gradient in station yards	Very flat to avoid movement of standing vehicles; need small slope for drainage	Max 1 in 400; min 1 in 1000

- **Ruling gradient** is not necessarily the steepest gradient; it governs the load of the engine for the whole section.
- **Vertical curves** (parabolic) are provided at changes of grade (when algebraic difference exceeds ≈ 0.4 %) with radius about 4000 m on main lines (IR practice).
- A gradient 'rising' or 'falling' in the direction of traffic; 'Pusher' gradients use banking engines.

2. Grade Compensation on Curves

On a curve, the **resistance to the motion of the train increases** (flange friction, wheel slip, lateral pressure). So the gradient on a curve is **reduced (flattened)** to keep the same tractive effort. This reduction is **grade compensation**.

Gauge	Compensation per degree of curve
Broad Gauge (BG)	0.04 % per degree
Metre Gauge (MG)	0.03 % per degree
Narrow Gauge (NG)	0.02 % per degree

$$\text{Compensated gradient} = \text{Ruling gradient (\%)} - 0.04 \times D$$

Example 1

BG, ruling gradient 1 in 150 = 0.667 %, curve of 4°.

Compensation = 0.04 × 4 = 0.16 %.

Compensated gradient = 0.667 − 0.16 = 0.507 % ≈ **1 in 197** (flatter than 1 in 150).

Example 2

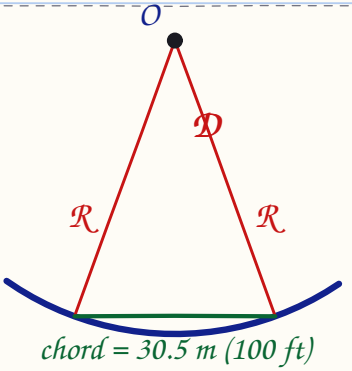
BG, ruling gradient 1 in 100 (1.0 %) and a 5° curve → compensation = 0.2 %.

Gradient on the curve = 1.0 − 0.2 = 0.8 % = **1 in 125**.

- Compensation is applied only where the curve lies on the gradient; **no compensation** is

needed if the gradient is already flatter than the compensated value.

3. Degree of Curve and Radius



Degree of curve (D) :

angle subtended at centre by a chord of 30.5 m length.

$R = 1720 / D$ (metres)

$D = 1720 / R$ (degrees)

Fig: Degree of a curve

D (degree)	1°	2°	3°	4°	5°	6°	8°	10°
R (metres)	1720	860	573	430	344	287	215	172

- Max degree of curve (IR): **BG 10°** ($R \approx 175$ m), MG 16°, NG 40°. In yards, curves are flatter.
- Types of curves: **simple, compound, reverse** curves; **transition curves** connect a straight to a circular curve.

Transition curve

A transition curve (cubic parabola / clothoid) gives a **gradual change** of curvature ($\infty \rightarrow R$) and of cant (0 → full), so the centrifugal force builds up smoothly.

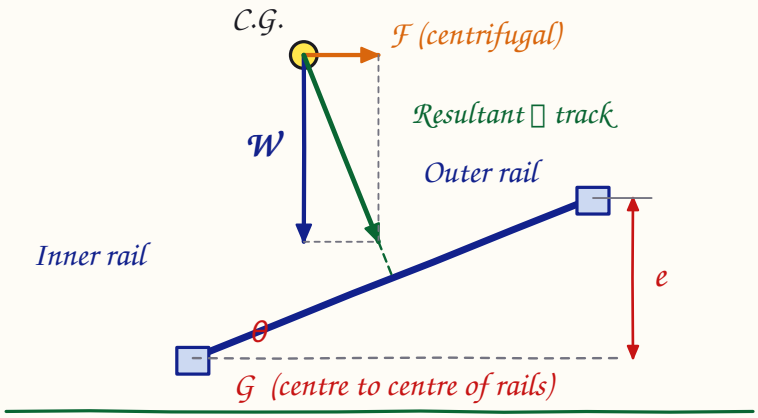
$$L = 0.008\ Ca\ Vm\ ;\ L = 0.008\ Cd\ Vm\ ;\ L = 0.72\ Ca$$

L in m, Ca = actual cant (mm), Cd = cant deficiency (mm), Vm = max speed (km/h). **Take the longest of the three.** Ex: $Ca = 100$, $Vm = 90 \rightarrow L = 0.008 \times 100 \times 90 = 72$ m and $0.72 \times 100 = 72$ m.

4. Cant (Super-elevation)

Definition

Cant (super-elevation) is the raised level of the outer rail over the inner rail on a curve. It is expressed in mm. It balances the centrifugal force $F = mV^2/R$ so that the resultant of W and F is perpendicular to the track.



$\tan\theta = e/G = F/W = V^2/(gR)$

$e = G V^2/(gR)$

In IR units : $e = G V^2/(127 R)$

e (mm), G = 1750 mm (BG), V (km/h), R (m)

$\Rightarrow e \approx 13.78 V^2/R$ mm (BG)

Outer rail raised → balances F

Fig: Cant on a curved track

Need for cant

- To counteract centrifugal force and keep the load shared equally on both rails.
- Prevents overturning of vehicle, side wear of rails and wheel flanges.
- Gives passenger comfort, reduces lateral thrust on track, and gives safe high speed.

Equilibrium cant

$$e = G V^2/(127 R) \quad [mm]$$

G = 1750 mm for BG (centre-to-centre of rails), V in km/h, R in m. So $e = 13.78 V^2/R$ (BG). For MG G ≈ 1058 mm.

- Maximum cant: BG 165 mm (185 mm on some Group A/B routes), MG 90 mm, NG 65 mm; also should not exceed G/10.
- Cant is gradually run out on the transition: cant gradient 1 in 720 (desirable), not steeper than 1 in 360; rate of change ≈ 35 mm/s.
- Equilibrium speed is chosen between goods and fast trains (≈ 3/4 of max speed for mixed traffic).

Example 3

BG curve, $R = 600$ m, equilibrium speed $V = 80$ km/h.

$e = 13.78 \times 80^2 / 600 = 13.78 \times 6400 / 600 =$ **147 mm** (< 165 mm, OK).

Example 4 – speed for a given cant

$R = 600$ m, actual cant given = 100 mm, find equilibrium speed.

$V = 0.27\sqrt{(e\,R)} = 0.27\sqrt{(100 \times 600)} = 0.27 \times 244.9 =$ **66 km/h**.

5. Cant Deficiency and Cant Excess

Cant deficiency (Cd)

The **difference between the equilibrium cant** required for a speed and the **actual cant provided**. Allowed when train speed is higher than equilibrium speed and the rider sits a little 'un-banked'.

- Maximum **cant deficiency**: BG **75 mm** (100 mm on selected high-speed routes), MG 50 mm, NG 40 mm.
- **Cant excess** (Ce) = actual cant − equilibrium cant for the speed of a slow train; max 75 mm; excess causes **inner rail wear**.

$$\text{Equilibrium cant for } V_{max}: \; Ca + Cd = 13.78\, V_{max}^2 / R$$

Negative super-elevation (negative cant)

On a **turn-out of contrary flexure**, the main line is on a curve of cant e, but the branch curves the opposite way. Since both tracks are at the same level, the branch gets the **outer rail lower** than the inner rail – this is **negative cant (Ne)** equal to the cant of the main line there.

- It increases the effective deficiency, so **speed on the branch is restricted**.

$$V = 0.27\sqrt{[\,(Cd(max) - Ne) \times R\,]}$$

Example 5

Branch $R = 300$ m, $Cd(max) = 75$ mm, negative cant $Ne = 20$ mm.

$V = 0.27\sqrt{[(75 - 20) \times 300]} = 0.27 \times 128.4 =$ **34.7 ≈ 34 km/h**.

6. Safe Speed on Railway Track

Safe speed on a curve depends on radius, cant, cant deficiency, length of transition and type of track. The lowest of the speeds below is adopted.

Method	Formula (V km/h)
(i) Cant + deficiency, transitioned curve	$V = 0.27\sqrt{[(Ca + Cd)\,R]}$
(ii) Non-transitioned curve (BG)	$V = 0.27\sqrt{[(Ca + Cd - 75)\,R]}$
(iii) Martin's formula – BG & MG, transitioned	$V = 4.35\sqrt{(R - 67)}$
(iv) Martin's formula – NG / non-transitioned (×0.8 rule)	$V = 3.65\sqrt{(R - 6)}\,(NG)$
(v) Limiting by transition length	from $L = 0.008\,Ca\,V$ (or $Cd\,V$), $V = L/(0.008\,Cd)$

Example 6

BG, $R = 600$ m, $Ca = 100$ mm, $Cd = 75$ mm, transition provided.

(i) $V = 0.27\sqrt{(175 \times 600)} = 0.27 \times 324 = 87.5$ km/h

(iii) $V = 4.35\sqrt{(600 - 67)} = 4.35 \times 23.09 = 100.4$ km/h

Safe speed = lower value ≈ 87 km/h.

- Speeds are also limited by track structure, signals, gradient, vehicle design and sight distance.

7. Points and Crossings

Points (switches) and crossings are arrangements which allow a train to change from one track to another. A turn-out = points + lead rails + crossing.

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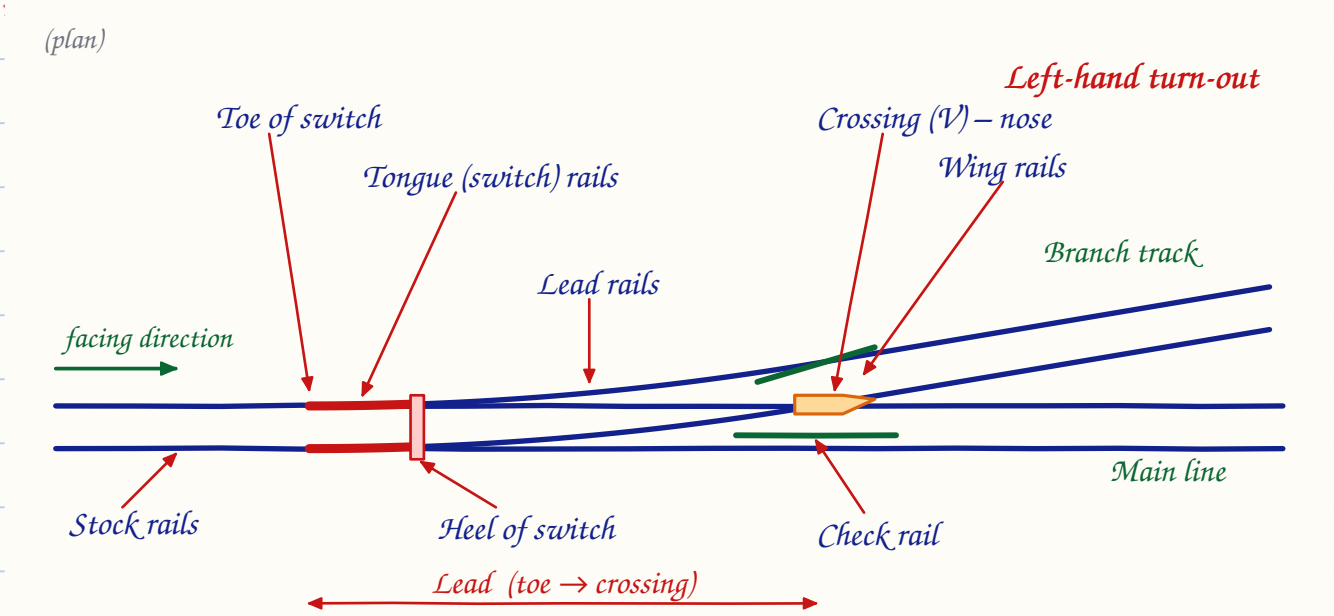


Fig: Layout of a left-hand turn-out

Components

Part	Description
Stock rail	Fixed outer rail against which the tongue rail fits; 2 stock rails (one on each side)
Tongue (switch) rail	Movable tapered rail, pivoted at heel; 2 per switch; moved by lever through stretcher bar; throw ≈ 100-115 mm
Toe / heel of switch	Toe = free thin end of tongue; heel = fixed end (heel block)
Lead rails	Curved rails connecting heel of switch and crossing
Crossing	V-shaped: nose (point rail), wing rails; lets wheel flange cross the other rail; wheel gap gets jumped
Check rails	Placed opposite the nose to guide wheel flange and prevent it from hitting the nose

Number of crossing and angle

Number of a crossing N = 1 in N: the spread is 1 unit for N units of length. Right-angle method: $N = \cot\alpha$; centre-line method: $N = \frac{1}{2}\cot(\alpha/2)$.

Crossing	1 in 8½	1 in 12	1 in 16
Angle α	6° 42′ 35″	4° 45′ 49″	3° 34′ 35″

Crossing	1 in 8½	1 in 12	1 in 16
Use	Yards, slow turnout (≈15 km/h)	Main line turnout (≈30 km/h; up to 50 with curved switch)	Higher-speed turnout

Left hand and right hand turn-outs

- **Left-hand (LH) turn-out** – when a person stands at the **toe** looking towards the crossing (facing direction), the branch diverges to the **left** of the main line.
- **Right-hand (RH) turn-out** – the branch diverges to the **right**. Layout is mirror image; the crossing/ switch parts are the same.
- **Facing point**: train meets toe first; **trailing point**: train enters from the heel / crossing side.

Functioning

- Lever (or point machine) moves the **stretcher bar** which shifts both tongue rails together by the throw.
- For the **main line**: left tongue closes on the left stock rail, right tongue opens; for the **branch**: the other way round.
- Points are **locked** (lock bar / clamp) and **detected** electrically and interlocked with signals so a signal clears only if points are set properly.
- Wheel flange follows the closed tongue, crosses at the crossing guided by check rails, and moves to the diverging track.

Other arrangements

Layout	Description
Crossover	Two turn-outs connecting two parallel tracks (trailing / facing)
Scissors crossover	Two crossovers superimposed, used in confined station yards
Diamond crossing	Two tracks cross each other at an angle (obtuse + acute crossings)
Single / double slip	Diamond + switches inside; allows movement from one track to the other
Triangle	Three turn-outs to reverse a train or engine (wye)

8. Station Yards

A station is a place where trains stop for passenger/goods handling or crossing. A yard is a system of tracks (sidings) laid within defined limits for receiving, sorting, storing and despatching vehicles.

Classification of stations	Description
Halt / flag station	Minor wayside station where only passengers get on/off; trains stop only if needed (flag)
Crossing / Block station	Station where two trains cross or overtake; has loop lines; equipped for block working
Junction	Where two or more lines meet/diverge; Large yards and platforms
Terminal station	End of the line (dead-end or through-terminal); e.g. CSMT, Howrah

Types of yards

- Passenger yard – receives and despatches passenger trains; platforms, coaching depot, washing lines.
- Goods yard – loading/unloading sheds, platforms, cranes, weighbridge.
- Marshalling yard – wagons are sorted, classified and made into new trains. Types: flat yard (shunting by engine), gravity yard, hump yard (wagons run down by gravity after being pushed over the hump).
- Locomotive yard – engine sheds, turn-tables, fuel/water facilities, repair workshops.

Requirements of a good station yard

- On a straight and level (or very flat) track; adequate length and space for expansion.
- Located so as to serve the traffic area, with access roads and proper drainage and signalling.

9. Signalling

Signals are devices that convey instructions to the loco pilot regarding movement of a train, to ensure **safe, quick and orderly** running, to keep trains apart (block system), and to control movement at stations and junctions.

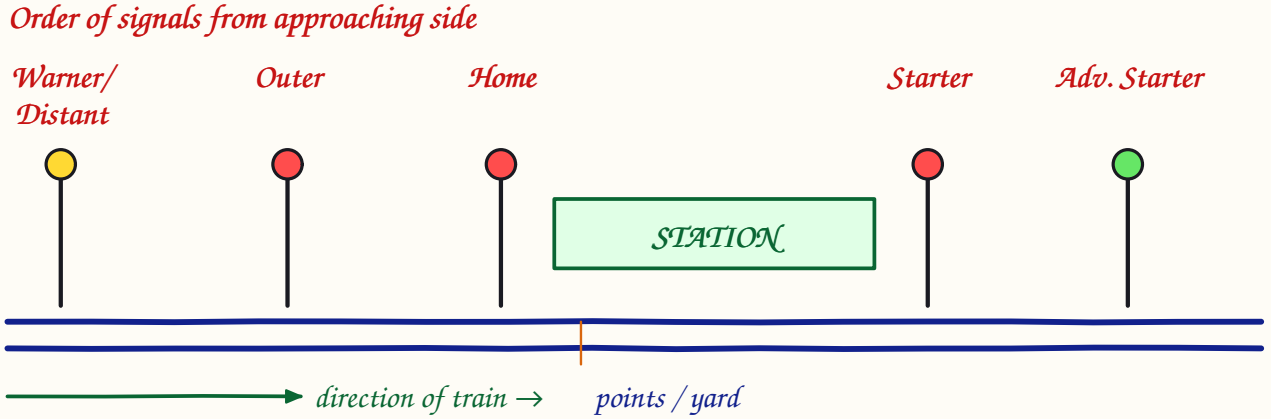


Fig: Fixed signals on a station approach

Classification of signals

Basis	Types
Function	Stop signals (Outer, Home, Starter, Advanced starter, Routing), Warner/Distant signal, Shunting signal
Operation	Mechanical (semaphore arm & wire), Electrical (colour-light, electrically operated)
Mode of display	Semaphore (arm, lower/upper quadrant), Colour-light (2, 3 or 4 aspects)
Method of use	Fixed signals, hand signals (flags/lamps), detonators (audible; for fog & emergency)
Location / Position	Warner/Distant → Outer → Home → Starter → Advanced Starter

Functions of fixed signals

- **Warner / Distant** – warns the driver to be ready to stop at the next stop signal; shows caution (yellow); located about 1 km (braking distance) before.
- **Outer** – first stop signal of a station; protects the station; stands at a safe distance outside Home.
- **Home** – protects station limits / points; driver can enter the platform line only if it is

OFF; stands ≈ 180 m before the first set of points.

- Starter – at the departure end of the platform; authorises train to leave. Advanced starter – last stop signal, at the end of the yard.
- Colour-light aspects: Red = STOP; Yellow = caution (be ready to stop at the next signal); Double yellow = preliminary caution; Green = proceed at full speed.
- Block system: the line is divided into blocks; only one train is allowed in a block at a time (absolute block, automatic block). Modern train protection: Kavach (ATP).

10. Interlocking

Definition

Interlocking is an arrangement of signals, points and other equipment, inter-connected in such a way that their movements are done in a fixed sequence, so that conflicting movements are impossible and a signal can be cleared only when the route is safe.

Objectives / principles

- A signal can be taken 'OFF' only if all points are correctly set and locked for the route and conflicting signals are 'ON'.
- Once signal is OFF, the points cannot be moved (locked) until the signal is put back to 'ON' and the train has passed.
- Two conflicting signals cannot both be OFF; facing points must be locked; the route must be isolated/free (track circuit).

Methods of interlocking

Method	Description
Mechanical (tappet & lock, key)	Levers in a frame with tappets and locking bars; Key interlocking uses transferable keys; Lock-and-block
Electrical / Relay interlocking	Relays and circuits control signals and points; operated from a control panel (route relay interlocking, RRI)
Solid State Interlocking (SSI)	Electronic microprocessor modules replace relays; compact, more reliable, faster

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Method	Description
Computer-based (EI)	Electronic interlocking with software, centralised traffic control (CTC)

- Interlocking has **Standards I to IV** (higher the standard more the safeguards, used for higher speeds and busier stations).
- Benefits: **safety**, increased line capacity, ease of operation, reduced human errors.

Important Questions – Unit II

Short answer (2 marks)

- What is ruling gradient? Define grade compensation.
- Define cant, cant deficiency and negative super-elevation.
- What is degree of a curve? Write the relation with radius.
- Define turn-out. What is the number of a crossing?
- Write any two functions of signals and what is interlocking?

Long answer (10 marks)

- Explain different types of gradients used on Indian Railways, and grade compensation with an example.
- Derive the formula for equilibrium cant; discuss cant deficiency and negative superelevation.
- Explain the safe speed on curves; solve a numerical on BG.
- With a neat sketch explain the layout of a left-hand turn-out and its functioning.
- Explain station yards, signalling and interlocking.

Quick revision: $e = GV^2/127R = 13.78V^2/R$ (mm) | $V = 0.27\sqrt{[(Ca + Cd)R]}$ | Max Cd BG 75 mm | Grade comp. BG 0.04 %/° | $R = 1720/D$ | $L = 0.008CaVm$.